CS 2XY3 - Problem Set 1: Processes and Scheduling

Due: Week 3. Submit via your LMS.

Question 1 (15 points)

Explain the difference between a process and a thread. Give two advantages of using threads over creating multiple processes for a web server handling concurrent requests.

Question 2 (20 points)

Consider the following set of processes with their arrival times and CPU burst lengths:

Process P1: Arrival=0, Burst=8ms
Process P2: Arrival=1, Burst=4ms
Process P3: Arrival=2, Burst=9ms
Process P4: Arrival=3, Burst=5ms

(a) Draw Gantt charts for FCFS, SJF (non-preemptive), and Round Robin (quantum=3ms) scheduling.

(b) Calculate the average waiting time and average turnaround time for each algorithm.

(c) Which algorithm gives the best average waiting time for this workload? Explain why.

Question 3 (15 points)

What is the convoy effect in FCFS scheduling? Provide a concrete example with 4 processes that demonstrates this problem, and explain how SJF avoids it.